

Scientists identified two distinct pathogens in blood samples taken from mice. They analyzed both pathogens in the laboratory and recorded the presence or absence of certain characteristics in the table below (NOTE: + indicates presence and – indicates absence).

Characteristics	Pathogen A	Pathogen B
Outer cell wall	–	+
Phospholipid membrane	+	+
Nucleus	–	–
DNA genome	+	+
Membrane-bound organelles	–	–
Ribosomes	–	+

They concluded that both pathogens are prokaryotic cells. Do the data support this conclusion?

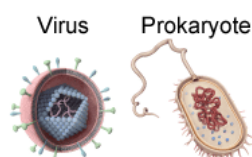
- ☐ A. Yes, because prokaryotes lack membrane-bound organelles. [49%]
- ✓ ☒ B. No, because ribosomes are found in all prokaryotes. [42%]
- ☐ C. Yes, because the genomes of all prokaryotes consist of DNA. [1%]
- ☐ D. No, because prokaryotes are not surrounded by an outer cell wall. [7%]

Correct

41% answered correctly

Explanation:

Characteristics of two pathogens



Characteristics	Pathogen A	Pathogen B
Outer cell wall	–	+
Phospholipid membrane	+	+
Nucleus	–	–
DNA genome	+	+
Membrane-bound organelles	–	–
Ribosomes	–	+

Pathogen A, which lacks ribosomes, cannot be a prokaryotic cell.

Disease-causing pathogens may be **classified** as **eukaryotic cells**, **prokaryotic cells**, or **viruses**. Viruses are unique in that they are obligate intracellular parasites that **rely on host cells to reproduce**. Accordingly, viruses **lack** the **transcriptional and translational** machinery found in all prokaryotic and eukaryotic cells. In addition, while the genomes of all prokaryotic and eukaryotic cells are always composed of DNA, viruses may contain either **DNA** or RNA (eg, **+ssRNA virus**, **retrovirus**) genomes.

In this scenario, researchers identified two distinct pathogens in blood samples taken from mice. After laboratory analysis, they recorded the presence or absence of certain characteristics in each pathogen. The data show that pathogen A **lacks ribosomes**, a characteristic that is *unique* to **viral pathogens**. Because *all* prokaryotic cells contain ribosomes, **pathogen A cannot be a prokaryotic cell** and is most likely a *virus*.

(Choice A) In contrast to eukaryotic cells, membrane-bound organelles are absent in *all* prokaryotic cells. However, *viruses* also lack membrane-bound organelles, so the absence of membrane-bound organelles alone is *not* sufficient to conclude that both pathogens are prokaryotic.

(Choice C) DNA genomes are found in all eukaryotic and prokaryotic cells, as well as in some viruses. Therefore, the presence of a DNA genome does *not* necessarily indicate that a pathogen is prokaryotic.

(Choice D) Generally, prokaryotic cells and some eukaryotic cells (eg, fungi and plant cells) are surrounded by a protective cell wall. Accordingly, the presence of a cell wall in pathogen B indicates that the pathogen could be either prokaryotic or eukaryotic. However, because pathogen A does *not* have a cell wall, it is unlikely to be a prokaryotic cell.

Educational objective:

In contrast to eukaryotic cells, prokaryotic cells and viruses lack membrane-bound organelles. However, viruses are unique in that they may possess DNA or RNA genomes and lack the transcriptional and translational machinery (eg, ribosomes) found in prokaryotic and eukaryotic cells.

Time spent:0 sec

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System:
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